

Bilateral equalization of synaptic output in olfactory glomeruli of *Xenopus* tadpoles

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This manuscript investigates inter-hemispheric interactions in the olfactory system of *Xenopus* tadpoles. Using a combination of electrophysiology, pharmacology, imaging, and uncaging, the transection of the contralateral nerve is shown to lead to larger odor responses in the unmanipulated hemisphere, and implicates dopamine signaling in this process. The study uses a rich and sophisticated array of tools to investigate olfactory coding and uncovers **valuable** mechanisms of signaling. However, the data is **incomplete**, with a few of the conclusions not being well-supported by the data; the interpretation should be adjusted with some caveats, or additional experiments should be done to support these conclusions.

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Abstract

Odorants stimulate olfactory sensory neurons (OSNs) to create a bilateral sensory map defined by a set of glomeruli present in the left and right olfactory bulbs. Using *Xenopus tropicalis* tadpoles we challenged the notion that glomerular activation is exclusively determined ipsilaterally. Glomerular responses evoked by unilateral stimulation were potentiated following transection of the contralateral olfactory nerve. The gain of function was observed as early as 2 hours after injury and faded away with a time constant of 4 days. Potentiation was mediated by the presence of larger and faster calcium transients driving glutamate release from OSN axon terminals. The cause was the reduction of the tonic presynaptic inhibition exerted by dopamine D₂ receptors. Inflammatory mediators generated by injury were not involved. These findings reveal the presence of a bilateral modulation of glomerular output driven by dopamine that compensates for imbalances in the number of operative OSNs present in the two olfactory epithelia. Considering that the constant turnover of OSNs is an evolutionary conserved feature of the olfactory system and determines the innervation of glomeruli, the compensatory mechanism here described may represent a general property of the vertebrate olfactory system to establish an odor map.

Introduction

Olfactory glomeruli are spherical regions of neuropil that contain the first synapse for processing olfactory information and form the glomerular layer of the olfactory bulb. Despite variations in number or size among vertebrate species, glomeruli show an evolutionarily conserved synaptic connectivity (Shepherd, Chen, and Greer 2004 [↗](#)). Glomerular input comes from axon terminals of olfactory sensory neurons (OSNs) expressing the same odorant receptor that transfer information to the dendrites of mitral cells. Synaptic transmission between OSNs and mitral cells is regulated by inhibitory and excitatory contacts established with juxtaglomerular neurons. This connectivity allows that upon exposure to odorants only a defined set of glomeruli is activated, creating an odor map (Mombaerts et al. 1996 [↗](#)). Glomeruli exhibit an anatomical and functional symmetrical distribution in the left and right olfactory bulbs, therefore, the sensory map created in the olfactory bulb is bilateral and is generated by homologous glomeruli reflecting the contribution of the two nasal cavities (Lodovichi 2021 [↗](#); Belluscio and Katz 2001 [↗](#)).

OSNs originate in the olfactory epithelium and are subject to a continuous turnover throughout life (Holl 2018 [↗](#)). An efficient neuronal replacement is accomplished by the rapid connection of newborn OSNs to the olfactory bulb. For example, in *X. tropicalis* tadpoles OSNs can establish functional synapses in a time window of 4 days (Terni et al. 2017 [↗](#)) and, in mice, it takes a week for newborn OSNs to get inserted in the olfactory bulb through a *plug-and-play* mechanism (Browne, Crespo, and Grubb 2022 [↗](#)). The balance existing between the elimination of OSNs and their insertion in olfactory bulb circuitry likely determines a range of input neurons innervating a glomerulus, rather than a constant, precise figure. In this context, how the formation of an odor map accounts for possible variations in the number of input neurons is unknown. Considering that OSNs exclusively form ipsilateral synapses (Shepherd, Chen, and Greer 2004 [↗](#)) and release glutamate with near-maximal release probability (Murphy et al. 2004 [↗](#)), the possible crosstalk balancing the individual contribution of bilaterally distributed glomeruli remains unknown, because their high output gain is assumed to be defined unilaterally.

Here, we take advantage of the experimental capacities offered by *Xenopus tropicalis* tadpoles to assess whether the output of a genetically labelled glomerulus (Terni et al. 2017 [↗](#); Terni and Llobet 2021 [↗](#)), is solely determined by ipsilateral stimulation. In vivo recordings carried out after disrupting the contribution of the contralateral pathway, challenged this notion. Our results show that the tonic inhibition of glomerular responses exerted by dopamine D₂ receptors is bilaterally shaped to compensate for differences in the number of OSNs innervating the contralateral olfactory bulb. Considering the evolutionary conserved developmental, morphological, and functional features of the *Xenopus* tadpole olfactory system (Menini 2010 [↗](#); Manzini, Schild, and Di Natale 2022 [↗](#)), the homeostatic mechanism here described might represent a general rule to the formation of an integrated odor map in vertebrates.

Materials and methods

Animals

Ethical procedures were approved by the regional government (Generalitat de Catalunya, experimental procedure #10753). *Xenopus tropicalis* were housed and raised according to the standard protocols of the animal facilities of the University of Barcelona. Tadpoles were obtained by natural mating and kept in tanks at 25 °C. *Xenopus* water conductivity was adjusted to 700 $\mu\text{S}\cdot\text{cm}^{-1}$, pH=7.5. Tadpoles at stage 47-52 of the Nieuwkoop–Faber criteria (Nieuwkoop and Faber 1956 [↗](#)) were used for the experiments. The *X. tropicalis* transgenic line *Dre.mxn1:GFP*

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To apply surgical procedures, *Xenopus* tadpoles were anesthetized in 0.02% MS-222 and transferred to a wet nitrocellulose filter paper placed under a stereomicroscope. Iridectomy scissors were used for the unilateral transection of an olfactory nerve or the bilateral transection of optic nerves. Tadpoles were returned to water tanks after surgery.

Electrophysiology

X. tropicalis tadpoles were anesthetized in 0.02% MS-222 and the portion of skin covering the olfactory bulb was removed. Animals were transferred to a well fabricated in a dish coated with silicone elastomer. A coverslip restricted tadpole movements and left olfactory placodes and bulbs accessible (Terni et al. 2018 [DOI](#)). The dish was placed on the stage of an upright microscope (Zeiss, Axioexaminer A1, Oberkochen, Germany) and was continuously perfused with *Xenopus* Ringer containing (in mM): 100 NaCl, 2 KCl, 1 CaCl_2 , 2 MgCl_2 , 10 glucose, 10 HEPES, 240 mOsm/kg, pH=7.8, supplemented with 1 μM d-tubocurarine to prevent muscle contractions. All salts and drugs were from Sigma-Aldrich (Saint Louis, MO). CGP-36742 was from Novartis Pharmaceuticals (Basel, Switzerland). To carry out extracellular recordings a borosilicate pipette filled with *Xenopus* ringer was targeted to the olfactory glomerulus showing GFP fluorescence. Signals were acquired using a Geneclamp 500A amplifier (Molecular Devices, San Jose, CA) and digitized at 10 kHz using a National Instruments NI-USB-6341 DAC board (National Instruments, Austin, TX) controlled by mafPC software (courtesy of M. A. Xu-Friedman, University at Buffalo, NY). The recorded changes of the LFP were low pass filtered below 100 Hz and analyzed with Igor Pro 9.0 (Wavemetrics, OR).

Methionine was chosen as odor stimulus based on its broad capacity to stimulate OSNs in *Xenopus laevis* tadpoles (Manzini and Schild 2004 [DOI](#)). A puff of 200 μM methionine solution, obtained by diluting a 10 mM stock solution prepared in *Xenopus* Ringer (pH=7.8), was applied on the ipsilateral olfactory epithelium to the recorded glomerulus. The amino acid was delivered through a 0.25 mm diameter fused silica capillary (World Precision Instruments, Hertfordshire, UK) positioned on the top of a nasal cavity. The timing of application was controlled via a TTL pulse. The characteristic odor evoked glomerular response was obtained by averaging LFP changes triggered by sequential stimulations applied at 2 min intervals. The local application of antagonists was carried out by transiently applying 20 psi pressure to the pipette holder inlet. Approximately 1 μL of the pipette solution was delivered to the glomerulus.

Imaging

Quantification of OSNs present in the olfactory epithelium of *Dre.mxn1:GFP* tadpoles was performed in an inverted LSM880 confocal microscope (Zeiss) using animals anesthetized in 0.02% MS-222. The number of neurons was estimated from maximal intensity projections in confocal Z-stacks. To selectively eliminate GFP positive OSNs the 2Phatal method (Hill et al. 2017 [DOI](#)) was adapted to confocal microscopy. Tadpoles were immersed for 15 minutes in *Xenopus* water containing 5 $\mu\text{g}/\text{mL}$ Hoechst 33342 and anesthetized in 0.02% MS-222. Animals were placed dorsally on a 25 mm glass #1.5 coverslip acting as the bottom of an imaging chamber and

transferred to the microscope stage. The right olfactory epithelium, located contralaterally to the recording site, was imaged with a 10X Plan Apo objective, NA=0.45 (Zeiss). The digital zoom was set to 3.5. Photobleaching was carried out using ZenBlue software (Zeiss) in 4 to 6 regions of interest (ROIs) containing two or more GFP positive OSNs. Animals were next returned to water tanks.

The *X. tropicalis* transgenic line *ElasGFP:Tubb2b-GCaMP6s*, where the neuronal β -tubulin promoter drives the expression of the calcium sensor GCaMP6s, was used to visualize the activation of olfactory glomeruli. As for in vivo electrophysiology, tadpoles were anesthetized in 0.02% MS-222, placed in a recording dish and transferred to the stage of an upright microscope (Zeiss, Axioexaminer A1). Tadpoles were continuously perfused with *Xenopus* Ringer containing 1 μ M d-tubocurarine. Imaging was carried out at 76 Hz using a Maico MEMS confocal unit (Hamamatsu Photonics, Hamamatsu City, Japan). A 40X, 0.75 NA water immersion W-N-Achroplan objective (Zeiss) was used. Odorant stimulation was performed as described for electrophysiology experiments. The timing of methionine application was synchronized with image acquisition (HCSImage software, Hamamatsu Photonics) using a Master 8 stimulator (AMPI, Jerusalem, Israel). In experiments where imaging was coupled to recordings of the LFP, the electrode was targeted to the lateral glomerular cluster. All signals were synchronized using a Master 8 stimulator.

Raw fluorescence image sequences were used to construct $\Delta F/F$ movies according to the relationship $(F-F_0)/F_0$, where F_0 corresponded to the basal fluorescence levels obtained during 1 s before stimulation. $\Delta F/F$ sequences were next subsampled by averaging groups of 10 frames to identify putative glomeruli, which were defined as 10-20 μ m diameter round structures consistently responding to sequential stimulations applied at 2 min intervals. The selected ROIs were transferred to the original raw fluorescence movie to quantify increases in basal intracellular calcium levels. A ROI showing $\Delta F/F$ increases evoked by methionine application to the ipsilateral olfactory epithelium that were ≥ 3 SDs above baseline levels was considered as a single glomerulus. The temporal response was quantified using Igor Pro 9 software (Wavemetrics, OR).

The production of ROS in the *X. laevis* HyPer-YFP was evaluated after transection of a single olfactory nerve. Tadpoles were imaged in an inverted confocal LSM 900 microscope (Zeiss) using a Plan-Apochromat 20X, 0.8NA objective. Hyper-YFP was excited at 405 nm and 488 nm. ROS levels were estimated from the relationship obtained between HyPer-YFP excited at 488 nm and 405 nm (Love et al. 2013 [\[3\]](#)).

Uncaging of Rubi-glutamate

Rubi-glutamate was added to the pipette solution at a final concentration of 300 μ M. Procedures were carried out considering the high sensibility to light of the caged compound (Fino et al. 2009 [\[4\]](#)). To prevent spontaneous activation of Rubi-glutamate, glass pipettes were coated with beeswax. The electrode was positioned above the lateral region of the glomerular layer using dim transmitted light. Next, the GFP labelled glomerulus was targeted using blue light attenuated with a ND filter (0.5 OD). Approximately 1 μ L of the pipette solution was injected in the glomerulus. Upon certifying that the recording of LFP signal was stable for 2 min, a TTL signal opened a shutter (Lambda SC, Sutter Instrument, Novato, CA) for 500 ms to deliver blue light (470 ± 20 nm) through the epifluorescence port. A diaphragm restricted light application to the region targeted by the electrode.

Statistical analysis

For statistical analysis, paired and unpaired t-test were used to evaluate differences between two experimental groups. Comparisons among three or more groups were performed using one way ANOVA followed by multiple comparison Tukey's HSD test. Average values are expressed as mean $\pm$ s.e.m. Statistical analysis was carried out using Igor Pro software.

Results

Characterization of odor evoked responses in a genetically defined glomerulus

The *Xenopus tropicalis* line *Dre.mxn1:GFP* allows the identification of an olfactory glomerulus located laterally and innervated by a discrete population of OSNs (Terni et al. 2017 [↗](#); Terni and Llobet 2021 [↗](#)). We targeted an electrode to the left GFP labelled glomerulus and recorded changes in the Local Field Potential (LFP) evoked by ipsilateral stimulation with an amino acid acting as a waterborne odorant (Figs. 1A [↗](#) and B [↗](#)). An olfactory response was triggered by 100 ms application of 200 μ M methionine to the olfactory epithelium, thus confirming the ability of *Xenopus* tadpoles to detect amino acids (Manzini et al. 2007 [↗](#)). Stimulation evoked a negative deflection of the LFP that resembled responses obtained in the glomerular layer of rats upon sniffing odors (Chaigneau et al. 2007 [↗](#)). A main difference between negativities recorded in *X. tropicalis* and those found in rodents is that the latter are respiration-locked, while in tadpoles there was a single deflection that recovered with a half time ranging from 1 to 4 s.

An estimate of the spatial extent of the LFP signal was obtained by evaluating responses after changing the position of the recording electrode. The characteristic profile of LFP changes disappeared when the pipette was directed to the mitral cell layer (Fig. 1C [↗](#)), thus illustrating they were confined to the glomerular layer. The consistent success rate (85%, n=269) obtained by targeting the GFP labelled glomerulus with the recording electrode decreased to 50% (n=36) when random locations linearly spaced by 50 μ m were tested within the glomerular layer. The example shown in Fig. 1D [↗](#) illustrates how the characteristic response to methionine was obtained only in one out of four positions tested. These observations could be explained by the capacity of the electrode to detect only activated glomeruli (Manzini et al. 2007 [↗](#)). The high success rate of the recordings guided by fluorescence support that the GFP labeled glomerulus responded to methionine. Considering many olfactory glomeruli in *Xenopus* tadpoles are broadly tuned (Manzini et al. 2007 [↗](#)), it is conceivable that the recorded glomerulus was also activated by other waterborne odorants, which were not tested in the current study.

The LFP signal likely sampled the entire volume of $15765 \pm 2119 \mu\text{m}^3$ (n=33) defined by fluorescence but, as the lateral cluster of glomeruli is markedly involved in the detection of amino acids (Weiss, Manzini, and Hassenklöver 2021 [↗](#)), the influence of adjacent glomeruli cannot be ruled out from the start. Considering the labelled glomerulus had a diameter of $\sim 30 \mu\text{m}$ (Fig. 1E [↗](#)) and there was a 50 μm axial resolution to obtain independent readouts (Fig. 1D [↗](#)), only a portion of surrounding glomeruli could theoretically participate in the observed LFP signal. The glomerular layer of the amphibian olfactory bulb is loosely organized compared to mammals, as glomerular units show different sizes and lack ensheathing astrocytes (Nezlin and Schild 2000 [↗](#); Gaudin and Gascuel 2005 [↗](#)). The absence of a homogeneous distribution of glomeruli, added to the dorsorostral location of GFP labelled axon terminals (Fig. 1E [↗](#)), restricted the number of adjacent glomeruli that could theoretically contribute to the recorded LFP signal to those found ventrally or caudally. Even if 50% of such units were activated by methionine, the decay of the LFP following a relationship inversely related to distance to the recording site, as reported for the mammalian olfactory bulb (Karnup et al. 2006 [↗](#)), would minimize their contribution. Altogether indicates that methionine application to the olfactory epithelium reliably activated the lateral glomerulus labeled by GFP in the *Dre.mxn1:GFP* line.

Glomerular responses were assayed by repetitive stimulation at two-minute intervals and all negativities showed a comparable profile (Fig. 1B [↗](#)). The peak of the negative deflection was reached ~ 1 s after olfactory stimulation. The characteristic LFP response of any given tadpole was obtained by averaging 5 to 8 consecutive stimulations. Local application of 100 μ M CNQX

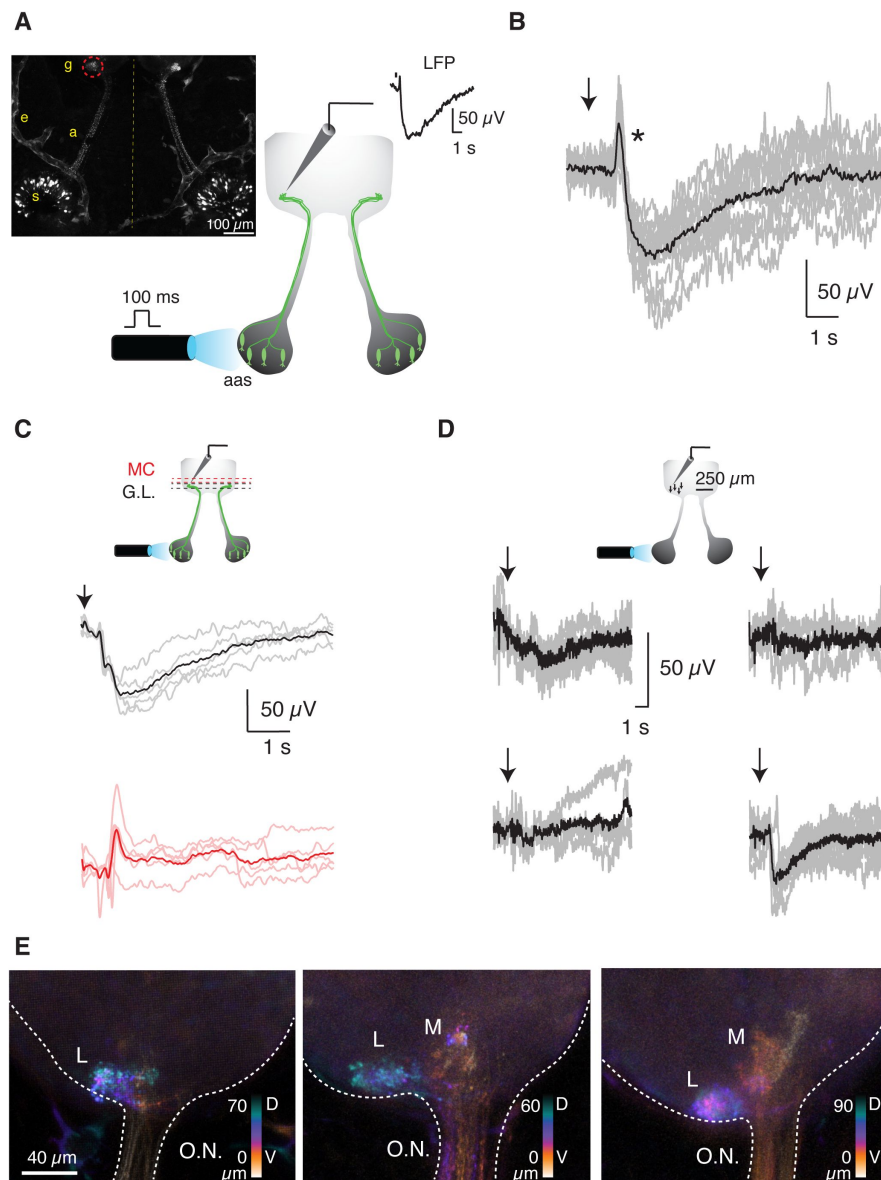


Figure 1.

Recording of odor evoked responses in a genetically defined glomerulus.

A) Schematic diagram illustrating the experimental preparation. A puff of 200 μM methionine was applied during 100 ms to the left olfactory epithelium of *Dre.mxn1:GFP Xenopus tropicalis* tadpoles. Changes in the Local Field Potential (LFP) were measured ipsilaterally with an electrode targeted to the lateral glomerulus formed by axon terminals of olfactory sensory neurons (OSNs) expressing GFP. The inset shows a confocal projection illustrating labelled OSNs (s: soma, a:axon) and the bilateral formation of glomeruli (g). The *Dre.mxn1* promoter also drives GFP expression in the endothelial cells of some blood vessels (e). **B)** Representative glomerular odor evoked response (black) obtained by averaging individual responses (gray) of the LFP following stimulation (arrow). The asterisk shows a positivity that appeared in 37% of the recordings and preceded the characteristic negativity associated to glomerular activation. **C)** Negative deflections of LFP (individual responses, gray; average trace, black) were observed when the recording electrode was placed in the GFP labelled glomerulus but disappeared in the mitral cell layer (MC, red traces). **D)** Experiment showing LFP recordings performed in four different locations of the glomerular layer spaced by 50 μm . The characteristic odor evoked response was observed in only one of the positions tested. The representative glomerular odor evoked response (black) was obtained by averaging individual responses (gray) following stimulation with methionine (arrow). **E)** Hyperstack projections of the left olfactory bulb of three different *Dre.mxn1:GFP Xenopus tropicalis* tadpoles. The lateral glomerular cluster (L) is always obvious, and some medial projections (M) are evident in two of the illustrated examples. Color scale indicates dorsoventral disposition.

decreased LFP negativities by approximately 50% (**Fig. 2A**). The reduction was comparable when 100 μ M D-AP5 was used instead of CNQX (**Fig. 2B**). The concomitant application of both drugs produced an additive effect that reduced responses by 70% (**Fig. 2C**, $p=0.0004$, paired t-test), thus evidencing their synaptic origin, as well as the involvement of AMPA and NMDA receptors. These results are comparable to those obtained in rats (Chaigneau et al. 2007; Lecoq, Tiret, and Charpak 2009), which validates our experimental configuration to record LFP responses in a region defined by a genetically labelled glomerulus. The onset phase of negativities was particularly sensitive to CNQX and AP5, suggesting that it was reflecting the involvement of receptors activated by glutamate release from OSN axon terminals. A way to test this possibility was to induce a jump in glutamate concentration inside the glomerulus and assay whether a change in the LFP signal occurred. We targeted the electrode to the lateral GFP glomerulus, injected *Xenopus* ringer solution containing 300 μ M Rubi-glutamate and uncaged it using a flash of blue light (Fino et al. 2009). A transient negativity was induced by a 500 ms pulse of light, which was reproduced by repeated stimulations (**Fig. 2D**). The average amplitude of LFP_{peaks} was of 30 ± 3 μ V ($n=21$, 3 tadpoles). This observation supported that the initial phase of LFP negativities was caused by the marked increase in the extracellular concentration of glutamate secreted by OSN axon terminals converging in a glomerulus.

The contribution of inhibitory neurotransmission mediated by GABA_A receptors was negligible because no variations in LFP changes evoked by methionine were observed after local application of 1 mM picrotoxin (**Fig. 2E**). To certify that ipsilateral stimulation of OSNs was the trigger of recorded responses, we investigated the effect of methionine application to the contralateral epithelium. Negativities disappeared (**Fig. 2F**), thus demonstrating that glomerular activation was exclusively unilateral. In 37% of the tadpoles studied ($n=174$) the negativity was preceded by a transient positivity (**Fig. 1B**). When present, this LFP deflection was not modified by glutamate and GABA_A receptor antagonists (**Figs. 2A** and **E**), thus indicating it was unrelated to synaptic mechanisms. As the positivity was caused by the application of olfactory stimuli and always occurred before the synaptic response, it is possible to consider its origin in cellular structures conveying peripheral information to the olfactory bulb such as the layer of nerve fibers.

Odor evoked responses depend on the number of input neurons

The amplitude of LFP negativities decreases during the reinnervation of the glomerular layer after olfactory nerve transection (Terni et al. 2017), which suggests that LFP changes are related to the number of OSNs axons projecting to glomeruli. If odor evoked responses were indeed related to glomerular input, the smallest negativities should take place in the earliest developmental stages. This possibility was confirmed by observing that GFP positive neurons found in the olfactory epithelium increased with development and their number followed an empirical linear relationship related to olfactory nerve width for tadpoles found between NF stages 47 and 54 (**Fig. 3A**, $r^2=0.96$). The peak of the LFP response was also related to olfactory nerve width but followed an exponential fit. Negativities reached a steady-state value (**Fig. 3A**), thus suggesting the achievement of a consolidated glomerular activation in this developmental time interval. The presence of constant glomerular responses for NF stages >50 coincides with the consolidation of the basic structure of the olfactory bulb, which remains constant through larval life (Byrd and Burd 1991). LFP changes were empirically described by the function $LFP_{peak} = LFP_{steady-state} + Ae^{-r \cdot ONW}$ that associated the maximum change in the LFP (in μ V) to olfactory nerve width (ONW, expressed in μ m). The fit revealed an increase rate (r) of $0.13 \mu\text{m}^{-1}$ starting at a diameter of 21 μ m, which corresponds to the thinnest olfactory nerves present during development around NF stage 40. Considering the linear correlation found between the number of GFP positive OSNs and the width of olfactory nerves, it was possible to rewrite the exponential fit as a function of the number of OSNs. The estimated peak negativity in the LFP could thus be obtained according to $LFP_{peak} = 108 - 1866 \cdot e^{-0.11 \cdot OSNs}$ for a given number of OSNs in the olfactory epithelium. On average, an olfactory epithelium contained 30 ± 2 GFP positive OSNs, ranging from 24 to 39 labelled neurons ($n=44$) and, this relationship provided a quantitative association of the amplitude of odor evoked responses to the insertion of OSNs in a glomerulus during normal development.

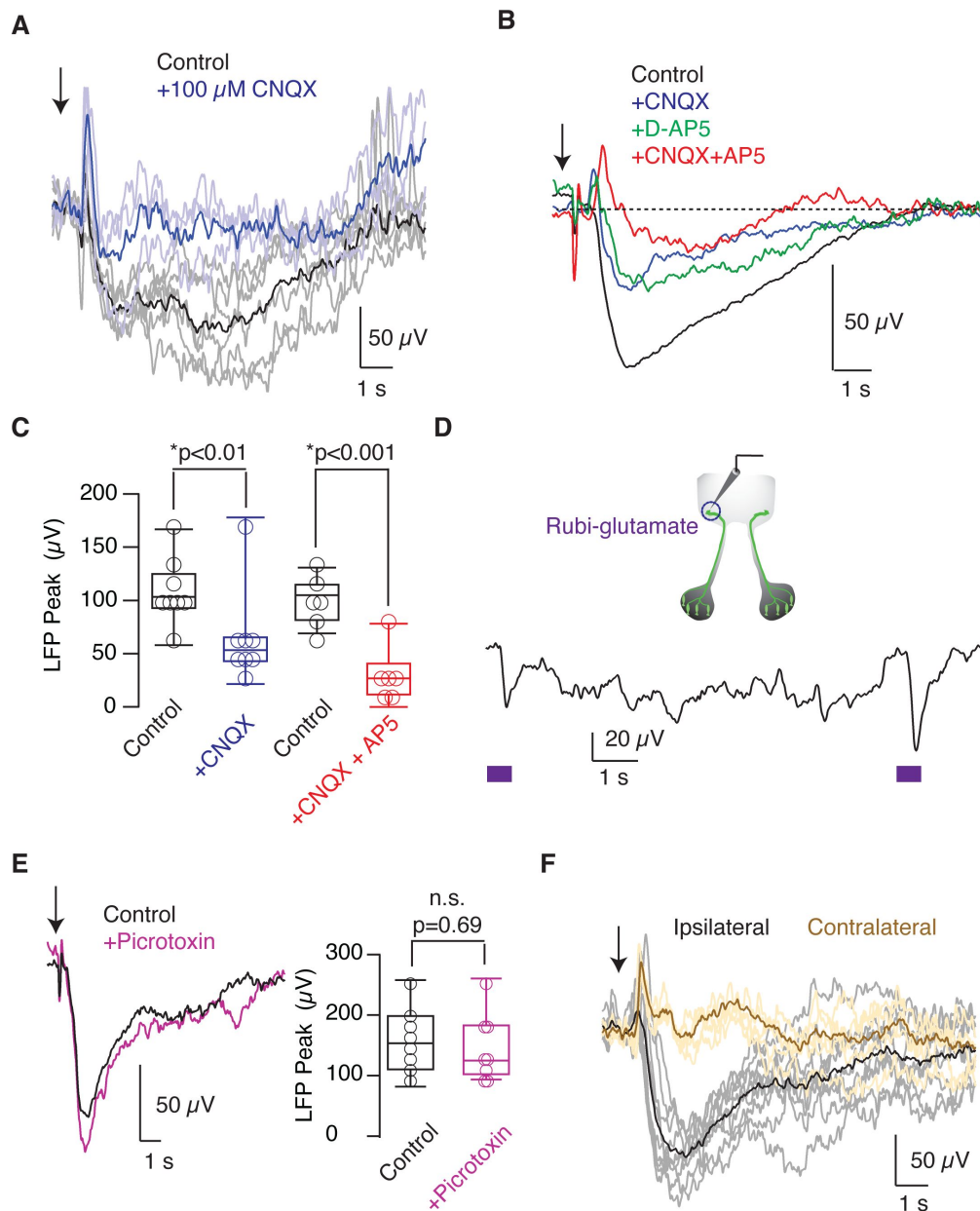


Figure 2.

Odor evoked responses are mediated by glutamatergic neurotransmission.

A) The representative glomerular odor evoked response (black) was obtained by averaging individual responses (gray) of the Local Field Potential (LFP) following stimulation (arrow). In this example, the pipette solution contained 100 μM CNQX and upon local injection of 1 μL , there was a reduction in the amplitude of LFP negativities. **B**) Mean LFP changes obtained under control conditions ($n=21$) were reduced after the application of 100 μM CNQX ($n=8$), 100 μM AP5 ($n=5$), or both 100 μM CNQX and 100 μM AP5 ($n=8$). **C**) Box plot illustrating how the initially recorded peak negativities were affected by the application of 100 μM CNQX, or, 100 μM AP5 together with 100 μM CNQX. Boxes represent the median (horizontal line), 25th to 75th quartiles, and ranges (whiskers) of the indicated experimental groups. Statistical differences were evaluated using paired t-test. **D**) Rubi-glutamate was injected in the GFP labelled glomerulus and locally uncaged with a 500 ms pulse of blue light (circle). The example shows the change in LFP induced by two flashes (squares) delivered at an interval of 9 seconds. **E**) Application of picrotoxin, a GABA_A antagonist, did not modify odor evoked changes. Recordings show average responses ($n=9$). Statistical differences were evaluated using paired t-test. **F**) Odor evoked LFP changes were exclusively triggered by ipsilateral stimulation. Individual responses are indicated in gray, with the representative average response in black. Contralateral stimuli (yellow) did not modify the LFP, as shown in the average representative response (brown).

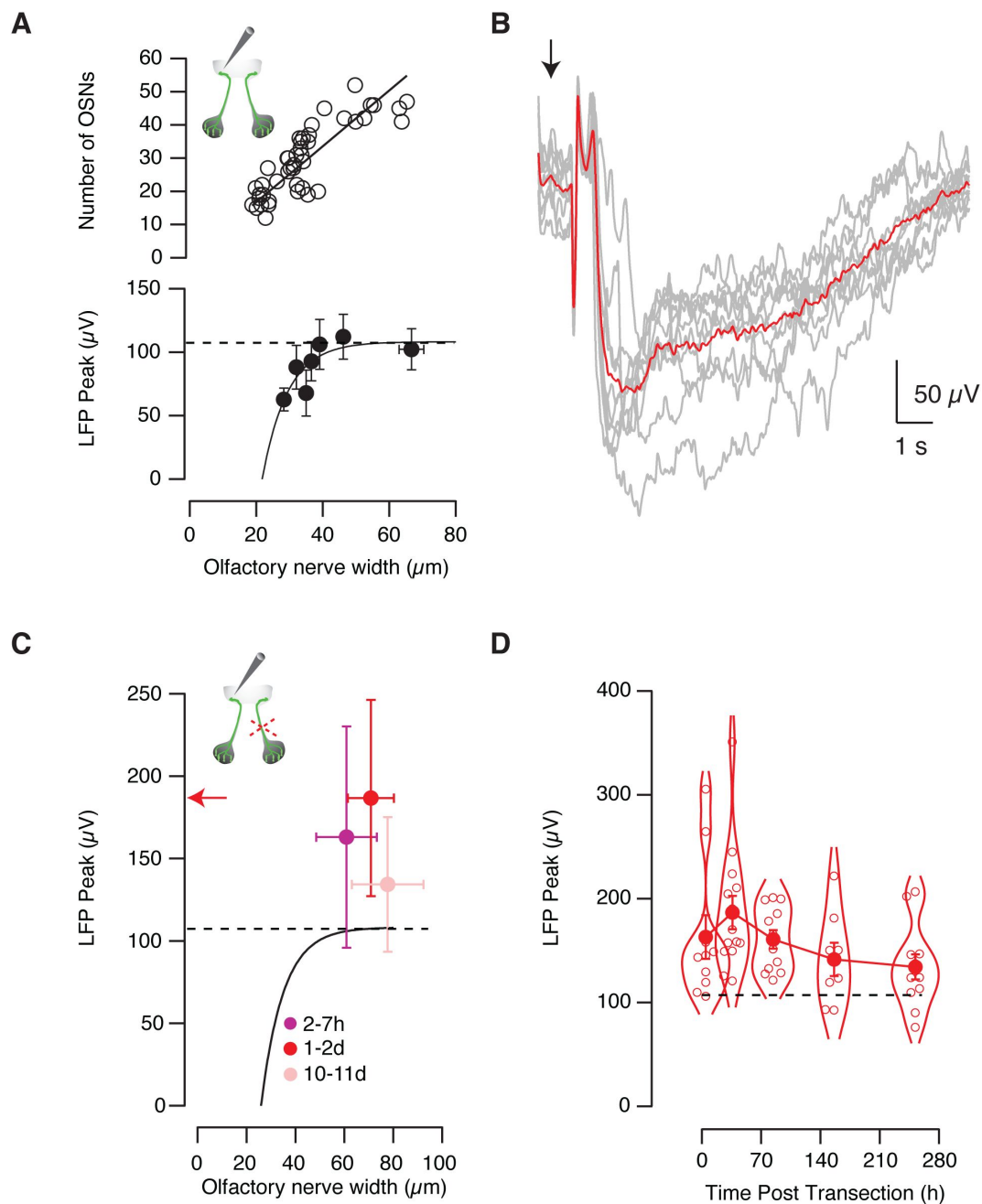


Figure 3.

Potentiation of odor evoked responses by transection of the contralateral olfactory nerve.

A) The number of olfactory sensory neurons (OSNs) and the amplitude of odor evoked negative deflections of the Local Field Potential (LFP) were related to olfactory nerve width according to linear and exponential functions, respectively. Individual data points are represented by circles ($n=48$). Each bin indicates the mean $\pm$ standard error of $n=6$ tadpoles. The dotted line indicates the steady-state LFP amplitude reached during development. **B)** Representative odor evoked response (red; individual responses, gray) obtained 24 h after contralateral nerve transection. A solution of 200 μM methionine was used as stimulus (arrow). **C)** Odor evoked LFP changes exhibit amplitudes above the expected values (dotted line as in A) after contralateral olfactory nerve transection at the indicated time points. The dots represent the mean $\pm$ s.e.m. obtained 2 to 7 hours ($n=10$), 1 to 2 days ($n=14$), and 10 to 11 days ($n=11$) post-injury. There was a 75% increase in animals recorded 1 to 2 days after transection of the contralateral olfactory nerve (red arrow) compared to control tadpoles (dotted line). **D)** Dots (mean $\pm$ s.e.m.) connected by a line illustrate odor evoked glomerular responses at the indicated times after injury. The superimposed violin plot displays individual data. Most LFP_{peak} values are above the level expected for the developmental period studied (dotted line as in A).

Odor evoked responses are potentiated by the injury of the contralateral olfactory pathway

Transection of both olfactory nerves causes a silencing of the capacity of *X. tropicalis* tadpoles to perceive odors (Terni et al. 2017 [DOI](#)). Since animals having a single nerve still display normal olfactory guided behavior, we investigated how information is transmitted in the absence of the mirror pathway. Olfactory nerve transection caused a potentiation of evoked LFP negativities in the recorded contralateral olfactory glomerulus (Fig. 3B [DOI](#)) that was already observed two hours after injury (Fig. C). All tadpoles investigated showed odor evoked responses above the expected LFP_{peak} level for the range of developmental stages analyzed. By reaching a maximum potentiation between 24 and 48 hours after transection, LFP_{peak} responses returned to control values following an exponentially recovery phase that took place with a time constant of 100 hours (Fig. 3D [DOI](#)). The overall duration of potentiation coincided with the approximately two weeks required to reform the glomerular layer upon olfactory nerve transection (Terni et al. 2017 [DOI](#)), suggesting that the increase of glomerular output occurred while the circuitry of the contralateral olfactory bulb was being remodeled after injury.

LFP_{peak} responses had a strong synaptic contribution because they were dependent on the number of OSNs innervating the glomerulus (Fig. 3A [DOI](#)), were sensitive to the application of glutamate receptor blockers (Figs. 2A-C [DOI](#)) and could be triggered by glutamate uncaging (Fig. 2D [DOI](#)). These results complement those obtained in rats showing a correlation of the amplitude of LFP negativities to calcium influx in OSN terminals (Lecoq, Tiret, and Charpak 2009 [DOI](#)) and EPSPs recorded in mitral cells (Chaigneau et al. 2007 [DOI](#)). We thus hypothesized that the observed potentiation affected to glutamatergic synapses mediating glomerular activation.

Tissue repair mechanisms are not involved in the potentiation of odor evoked responses

Since inflammatory mediators affect to synaptic functions (Wu et al. 2015 [DOI](#)), we evaluated the involvement of molecules released by injury using two different strategies. First, to observe if sensory nerve damage were enhancing LFP_{peak} responses, we sectioned both optic nerves (Fig. 4A [DOI](#)). Even though optic and olfactory nerves show substantial functional and anatomical differences, the nature of the injury, as well as the distance to the recorded site was comparable between optic and olfactory nerve transection. On these bases we assumed that the effect generated by diffusible inflammatory mediators, or the capacity to mobilize cells mediating inflammation, should be comparable in both experimental conditions. LFP negativities recorded 24h to 48h after bilateral optic nerve transection showed an amplitude of $86 \pm 6 \mu V$ ($n=17$). This value was expected by normal development but was significantly lower ($p=1.92 \cdot 10^{-5}$, unpaired t-test), than the amplitude of $186 \pm 16 \mu V$ ($n=14$) found in tadpoles subjected to transection of the contralateral olfactory nerve (Fig. 4B [DOI](#)).

Secondly, we investigated the release of reactive oxygen species (ROS), which are key to trigger tissue repair mechanisms in tadpoles (Love et al. 2013 [DOI](#)). The *Xenopus laevis* line HyPer-YFP displays a ubiquitous expression of the H_2O_2 sensor HyPerYFP that allows the in vivo detection of ROS created by wounds (Love et al. 2013 [DOI](#); Niethammer et al. 2009 [DOI](#)). Two hours after unilateral olfactory nerve transection there was an increase in H_2O_2 levels that was restricted to the cells present at the injury site (Fig. 4C [DOI](#)). ROS levels remained unaltered in the olfactory epithelium and in both olfactory bulbs. Moreover, the amplitude of odor evoked changes in the LFP were unaffected by the presence 200 μM apocynin or 2 μM diphenyleneiodonium, two blockers of ROS production (Fig. 4D [DOI](#), $p=0.36$, ANOVA followed by Tukey's test) that block tail regeneration at these concentrations (Love et al. 2013 [DOI](#)). All these findings together suggested that the

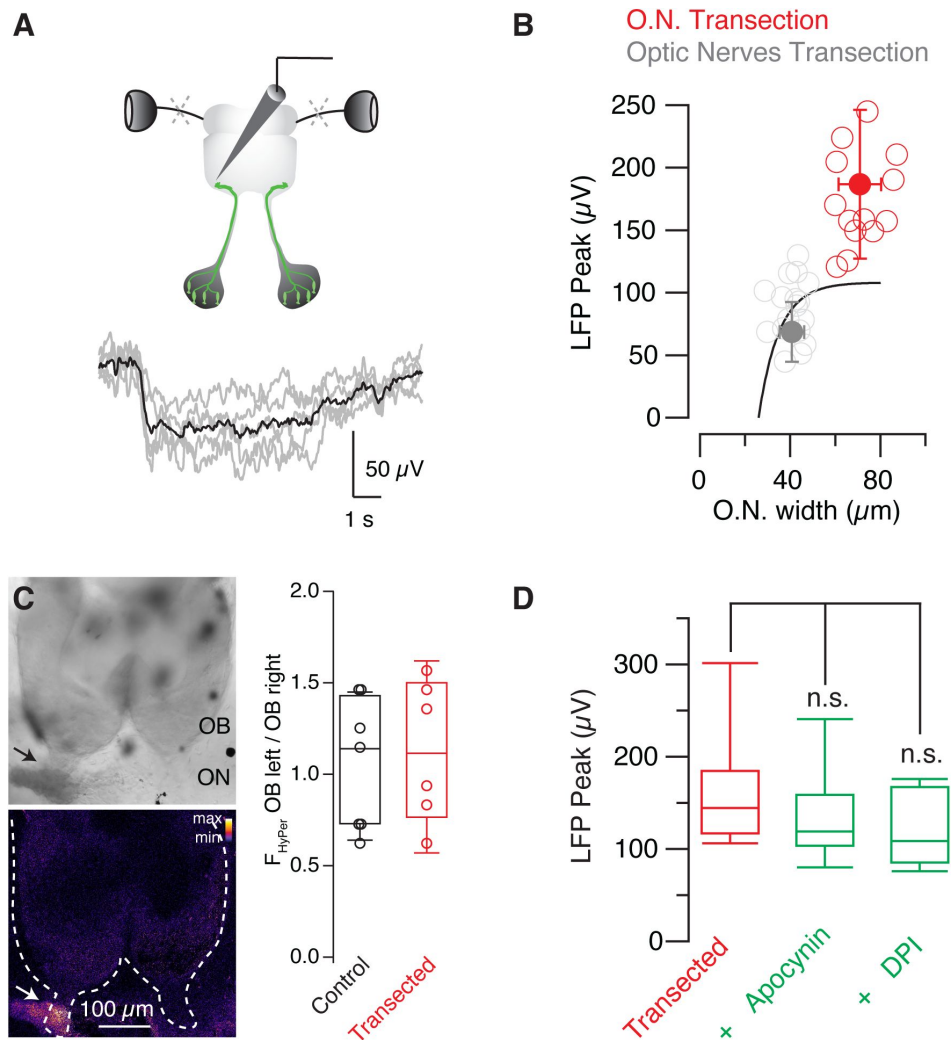


Figure 4.

The potentiation of odor evoked responses is not mediated by injury derived cues.

A) Odor evoked Local Field Potential (LFP) changes were recorded by an electrode targeted to the GFP-positive glomerulus of *Dre.mxn1:GFP* tadpoles one to two days after bilateral transection of optic nerves. **B)** Peak LFP negativities recorded in tadpoles with sectioned optic nerves did not exhibit the characteristic potentiation observed after transection of the contralateral olfactory nerve, as they remained within the range of values observed during normal development (solid line, as in **Fig. 3A**). Bins indicate mean $\pm$ s.e.m., circles show individual values. **C)** Imaging of reactive oxygen species (ROS) two hours after transecting one olfactory nerve (arrow). The ratio between the fluorescence emitted by HyPer-YFP when excited at 488 nm and 405 nm is indicated in pseudocolor. Notice that ROS were increased at the injury site but remained at basal levels in both olfactory bulbs as indicated by the box plot. Each circle shows values collected in a single tadpole. **D)** Block of ROS production by incubating tadpoles with 200 μ M apocynin ($n=10$) or 2 μ M diphenyleneiodonium (DPI, $n=5$) did not modify the amplitude of odor evoked LFP responses recorded 24 h after contralateral olfactory nerve transection ($n=10$). Boxes represent the median (horizontal line), 25th to 75th quartiles, and ranges (whiskers) of the indicated experimental groups.

enhancement of glomerular responses was unlikely caused by inflammatory mediators or the activation of fundamental tissue repair mechanisms operating in tadpoles and, revealed the presence of an intrinsic compensatory mechanism operating in the olfactory system.

Potential of odor evoked responses is of presynaptic origin

The activation of individual glomeruli was imaged in the transgenic *X. tropicalis* line *tubb2b:GCaMP6s* to gain insight on a putative synaptic mechanism acting on the bilateral balance of odor evoked responses. Ipsilateral application of 200 μ M methionine to the olfactory epithelium activated a defined set of glomeruli, which were identified as round, 10–20 μ m diameter structures. Same glomeruli reacted to three or more consecutive methionine stimulations (**Fig. 5A**). Responses detected as calcium transients showed comparable amplitudes and temporal profiles (**Fig. 5B**), suggesting that the simultaneous activation of sparse glomeruli contributed to the elaboration of an olfactory map. Two lines of evidence support that calcium increases originated in presynaptic terminals of OSNs. First, because the *tubb2b* promoter primarily targets OSNs over olfactory bulb interneurons. A detailed characterization carried out in *X. laevis* tadpoles shows that *tubb2b* efficiently drives expression of a genetically encoded fluorescent reporter in OSNs but fails to label periglomerular neurons positive for calretinin or tyrosine hydroxylase and only targets 24% of mitral cells (Daume et al. 2022). The convergence of multiple fluorescent OSN axon terminals on a single glomerulus and a minimal presence of labelled juxtglomerular and mitral cells, makes *X. tropicalis tubb2b:GCaMP6s* tadpoles well-suited to detect the presynaptic component of glomerular responses. Second, because the simultaneous imaging of glomerular activation and recording of LFP negativities revealed a temporal correlation of the two signals (**Fig. 5C**). This observation matches the findings of a previous study carried out in rats, where a comparable relationship was described between responses detected in OSN axon terminals labelled with a high affinity calcium dye and LFP signals (Lecoq, Tiret, and Charpak 2009). Although tadpoles of the *tubb2b-GCaMP6s* line consistently showed glomeruli activated by methionine, the coupling of imaging to LFP signals had a low success rate. Even though the electrode was targeted to the lateral glomerular cluster, which is the region expected to concentrate responses to amino acids (Weiss, Manzini, and Hassenklöver 2021), LFP negativities were detected in only 33% of animals tested ($n=17/52$). This figure is below the 50% found after the performance of recordings in random locations of the glomerular layer (**Fig. 1D**) but it is higher than the ~25% success rate found in *X. laevis* tadpoles (Manzini et al. 2007). This observation reinforces the selectivity of LFP recordings to detect the output of a single glomerulus, which in this set of experiments was coupled to the fluorescent response found in closest apposition to the tip of the pipette. Altogether indicates that the readouts of glomerular input and output were reported by transient increases in GCaMP6s fluorescence and LFP signals, respectively.

A hallmark of contralateral olfactory nerve transection was the development of faster LFP negativities. The time constant to reach the LFP_{peak} shortened significantly ($p=0.017$, unpaired t-test) from 1.05 ± 0.1 s ($n=46$ tadpoles) to 0.71 ± 0.1 s ($n=39$ tadpoles), whilst the recovery phase was unaffected (**Fig. 5D**, $p=0.79$, unpaired t-test). The faster glomerular output likely had a presynaptic origin, because calcium transients were larger and showed faster onset kinetics after contralateral nerve transection (**Figs. 5E** and **F**). The average amplitude increased more than two-fold ($p=0.02$, unpaired t-test) and the time constant describing the rise time shortened significantly ($p=0.0003$, unpaired t-test) in tadpoles with a transected olfactory nerve compared to control animals (**Fig. 5F**). The slow kinetics of GCaMP6s prevented a precise temporal association between calcium buildup and the onset of LFP changes (**Fig. 5C**) and a certain saturation of the high affinity calcium indicator in axon terminals was also expected. But, despite these limitations to quantitative analysis, the changes reported by GCaMP6s showed an overall enhancement of cytosolic calcium levels mediating neurotransmitter release. We next sought to investigate which regulatory mechanisms acting on OSN axon terminals were controlled bilaterally to alter glomerular output.

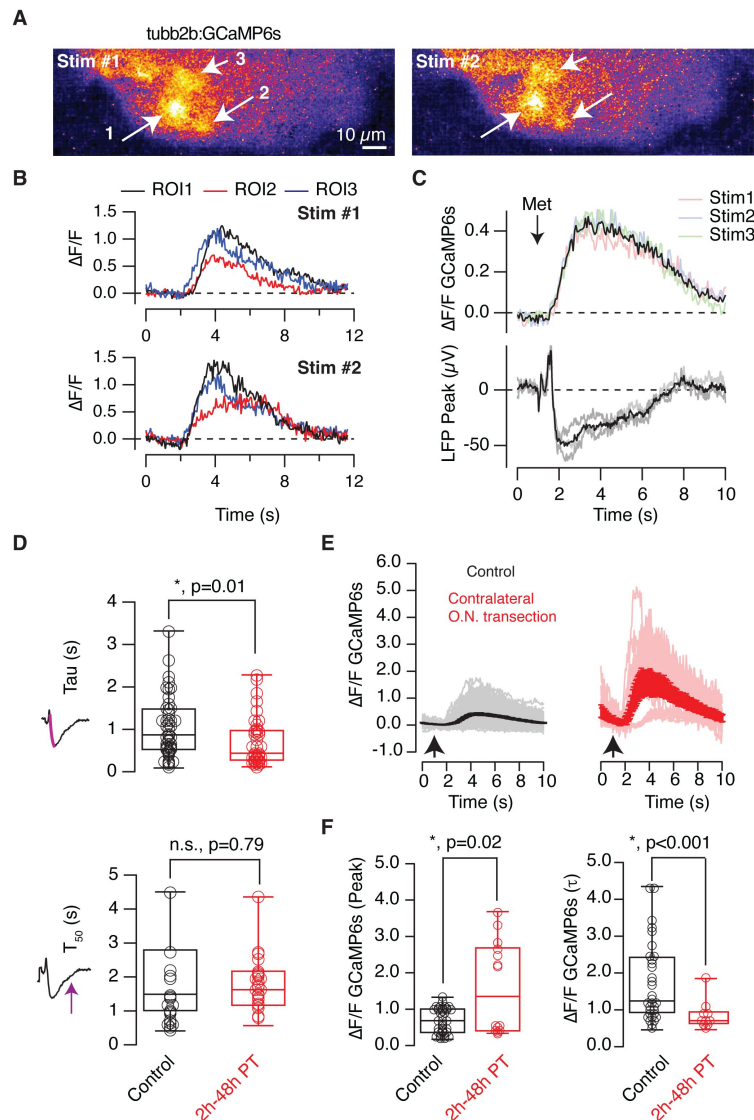


Figure 5.

The presynaptic component of glomerular activation is affected by damage to contralateral olfactory sensory neurons.

A) Application of a puff of 200 μ M methionine to the olfactory epithelium activates a set of glomeruli in the ipsilateral olfactory bulb (arrows) of *tubb2b:GCaMP6s* tadpoles. Images show the relative changes in GCaMP6s fluorescence ($\Delta F/F$) obtained after two sequential stimulations carried out in a single tadpole. **B)** Time course of the responses detected in the glomeruli indicated in A). **C)** Example showing the simultaneous recording of Local Field Potential (LFP) and changes in GCaMP6s fluorescence in the region targeted by the electrode. Colored traces and gray traces show the change in GCaMP6s fluorescence ($\Delta F/F$) and LFP respectively observed after three sequential applications of 200 μ M methionine. Black traces show the average $\Delta F/F$ and LFP responses. **D)** Kinetics of the change in LFP observed in tadpoles with the contralateral olfactory nerve transected between 2 h and 48h prior to recording. **E)** Intracellular calcium increases detected in glomeruli of control tadpoles with intact olfactory pathways (35 glomeruli, 10 tadpoles, black), and, in tadpoles subjected to the transection of the contralateral olfactory nerve (10 glomeruli, 3 tadpoles, red). Each trace indicates the response of a glomerulus to a single stimulus. Solid lines and error bars indicate mean $\pm$ s.e.m. **F)** Calcium transients detected in tadpoles with an olfactory nerve transected showed a larger amplitude and a rising phase with a shorter time constant (τ). Boxes in D) and F) represent the median (horizontal line), 25th to 75th quartiles, and ranges (whiskers) of the indicated experimental groups. Statistical differences in D) and F) were evaluated using paired and unpaired t-tests, respectively. Circles in D) indicate values obtained for a single tadpole and in F) refer to a single glomerulus.

Dopamine D₂ receptors modulated by contralateral glomerular input mediate presynaptic inhibition

Presynaptic inhibition regulates neurotransmission in olfactory glomeruli (McGann 2013 [↗](#)). Discrete populations of juxtaglomerular neurons that release GABA or dopamine activate GABA_B or D₂ receptors present in presynaptic terminals of OSNs to lower glutamate release (Wachowiak et al. 2005 [↗](#); Wachowiak and Cohen 1999 [↗](#)). Evidence points out that this is a tonically active mechanism (Pérez and Wachowiak 2008 [↗](#)), thus meaning the normal processing of odor information relies on a certain level of inhibition of glomerular output. To evaluate the possible involvement of dopamine or GABA in the potentiation of LFP_{peaks} observed after contralateral olfactory nerve transection, we locally applied antagonists of D₂ and GABA_B receptors to the recorded glomerulus. Odor evoked responses were enhanced by raclopride, a D₂ antagonist, in tadpoles with their olfactory pathways intact (Fig. 6A [↗](#)). The average negativity of 73±5 μV (n=8) experimented a gradual increase in the presence of 300 nM raclopride that reached 128±16 μV, 20 min after its application (Fig. 6B [↗](#), p=0.012, paired t-test). The potentiation of odor evoked responses caused by the D₂ antagonist was not observed in tadpoles with the contralateral olfactory nerve transected. Baseline negativities of 133±24 μV (n=11) were minimally increased by raclopride, changing to 156±33 μV (n=11, p=0.16, paired t-test), 20 min after drug application.

The time course of the increase in LFP_{peaks} observed in control tadpoles was well described by a Hill equation, reporting a time required to reach the 50% of the maximal increase of approximately 20 min (Fig. 6C [↗](#)). The time window describing the effect of raclopride could be attributed to disinhibition, since the primary target of presynaptic D₂ receptor activation is the decrease of cAMP levels and a consequent lowering of neurotransmitter release (Kaneko and Takahashi 2004 [↗](#)). The potentiation mediated by raclopride was comparable to the increase in LFP_{peaks} caused by olfactory nerve transection (Fig. 6B [↗](#), p=0.46, unpaired t-test). These results indicated that the tonic inhibition of glomerular output was comparably reduced either by directly inhibiting D₂ receptors, or, by surgically transecting the contralateral olfactory nerve. Further evidence for the involvement of D₂ receptors was obtained by simultaneously imaging GCaMP6s fluorescence and LFP signals (Fig. 6D [↗](#)). In the illustrated example raclopride shortened the time constant (τ) to reach the calcium peak from 1.3 s to 0.5 s and potentiated LFP changes by 50%, thus reproducing the effect of contralateral olfactory nerve transection on glomerular input and output, respectively (Figs. 3C [↗](#) and 5F [↗](#)). On average raclopride reduced τ from 1.27 s to 0.8 s (n=4 tadpoles, p=0.02, paired t-test), evidencing that the potentiation of glomerular responses was mediated by a faster buildup of presynaptic calcium levels that took place after antagonizing the inhibitory action of D₂ receptors.

Finally, we explored whether presynaptic GABA_B receptors activated by juxtaglomerular neurons (McGann 2013 [↗](#)) could mediate a synergic effect to D₂ receptors. This possibility was investigated by applying the selective GABA_B antagonist CGP-36742. LFP_{peaks} remained unchanged (Figs. 6E [↗](#) and F [↗](#), p=0.77, paired t-test). A significant tonic inhibition by GABA was thus ruled out, revealing dopamine was the main neurotransmitter inhibiting glomerular output in the glomerular layer of *Xenopus* tadpoles.

Effect on glomerular responses of the partial elimination of mirror olfactory sensory neurons

To shed light on the synaptic connectivity involved in the contralaterally driven potentiation of glomerular output we modified the 2Phatal technique (Hill et al. 2017 [↗](#)) to eliminate groups of GFP positive OSNs using a conventional confocal microscope (Fig. 7A [↗](#)). Tadpoles were placed during 15 minutes in *Xenopus* water containing 5 μg/mL Hoechst 33342, which is a membrane permeable dye with affinity for nucleic acids, to label all cells found within the olfactory epithelium. Damage was exerted by photobleaching groups of cells found in regions of interest

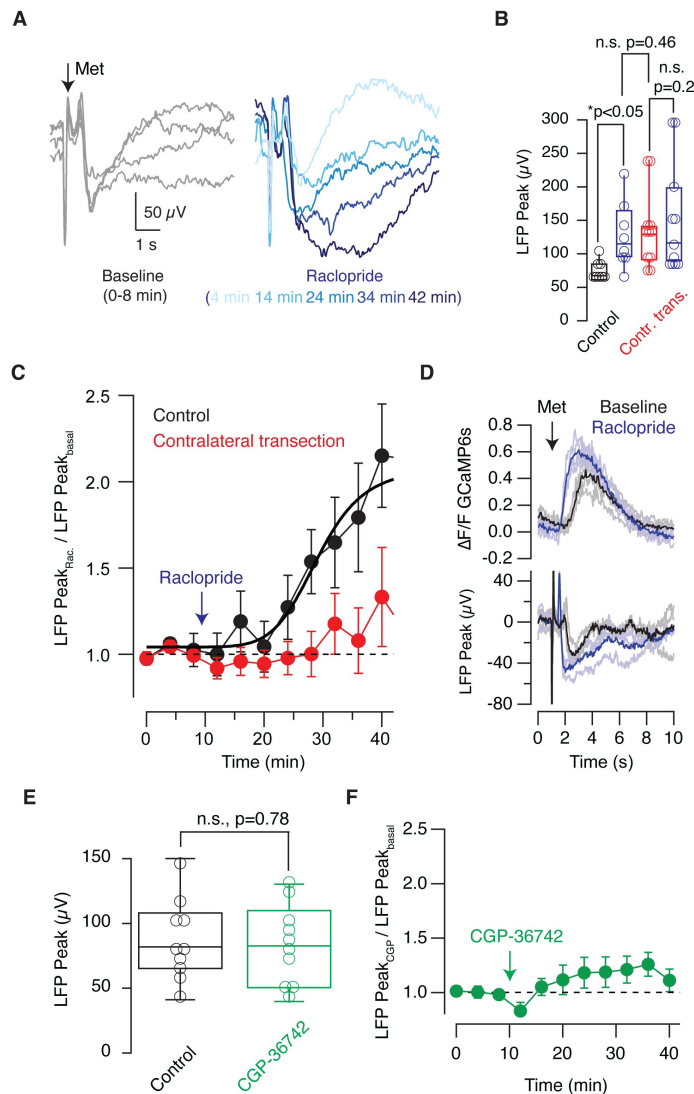


Figure 6.

Contralateral input modulates presynaptic inhibition mediated by dopamine D_2 receptors and is involved in the potentiation of glomerular responses.

A) Recordings obtained in a control tadpole showing how the amplitude of Local Field Potential (LFP) responses (gray traces) obtained during a baseline period of 8 minutes increase in a time-dependent manner after local application of 300 nM raclopride, a D_2 receptor antagonist. **B**) Box plot showing the effect of 300 nM raclopride (blue) on the amplitude of LFP responses recorded in tadpoles with full capacity to process odors (control, $n=8$, black) and tadpoles subjected to the transection of the contralateral olfactory nerve ($n=11$, red). Boxes represent the median (horizontal line), 25th to 75th quartiles, and ranges (whiskers) of the indicated experimental groups. The effect of raclopride was evaluated using the paired t-test and the comparison between control and transected groups was performed using the unpaired t-test. **C**) Relative change in LFP responses induced by 300 nM raclopride in control tadpoles and in tadpoles subjected to the transection of the contralateral olfactory nerve. Dots represent mean $\pm$ s.e.m. The solid black line illustrates the fit to a Hill equation, defining a T_{50} at 20 minutes. **D**) Simultaneous recording of LFP and changes in GCaMP6s fluorescence in a *tubb2b:GCaMP6s* tadpole. Gray traces and light blue traces show individual responses to sequential stimulations before and after application of 300 nM raclopride, respectively. Average responses are shown in black and dark blue. **E**) Application of CGP-36742, a $GABA_B$ receptor antagonist, did not modify LFP responses. The box plot compares the amplitude of LFP changes recorded before (gray) and 20 min after local application of 300 μM CGP-36742 (green). Statistical differences were evaluated using paired t-test. **F**) Time course of relative LFP changes induced by 300 μM CGP-36742. Dots represent mean $\pm$ s.e.m ($n=10$).

(ROIs) that contained two or more GFP positive OSNs. Efficient photobleaching of the nuclear label caused cell death, which was certified by the observation of condensed nuclei after 24 hours (**Fig. 7A**). The use of the confocal microscope precluded the achievement of single cell ablation but supported the robust elimination of cells within ROIs. Olfactory epithelial cells found outside the selected areas remained unaffected. Photobleaching was typically carried out in 4 to 6 ROIs, thus resulting in the elimination of 10 to 15 OSNs. Considering an epithelium contained on average ~30 GFP positive OSNs, the protocol led to an estimated 30 to 50% reduction of OSNs innervating the right GFP labeled glomerulus. Since the viability of tadpoles subjected to a more extensive reduction of fluorescent OSNs was compromised, all animals investigated still had a significant number of functional GFP labelled OSNs after photoablation.

A characteristic of the recorded odor evoked glomerular responses was their reproducibility. For a given tadpole, LFP negativities occurred after successive stimulations with a comparable profile (**Figs. 1B**, **2A**, **4A**), and showed an average variance in their amplitude of $386 \pm 96 \mu V^2$ ($n=17$). Variance of the responses obtained increased more than three-fold in tadpoles subjected to transection of the olfactory nerve (see for example **Fig. 3B**), reaching an average value of $1488 \pm 280 \mu V^2$ ($n=14$). An enhancement of variance was thus associated to the potentiation observed in response to olfactory nerve transection. Incubation of tadpoles with Hoechst 33342 neither modified the amplitude nor the variance of LFP_{peaks} (**Fig. 7B**). LFP negativities in tadpoles presenting a decimated population of contralateral GFP positive OSNs showed an increase in LFP_{peaks} from $80 \pm 5 \mu V$ ($n=17$) to $89 \pm 7 \mu V$ ($n=10$, $p=0.32$, unpaired t-test) and in variance to $1024 \pm 475 \mu V^2$ ($n=10$, $p=0.21$, unpaired t-test).

The non-significant changes in LFP_{peaks} amplitude and variance induced by selective photoablation could be attributed to the limited suppression of GFP labeled OSNs and it was not possible to establish through this experimental approach the effect of the complete suppression of cognate neurons. LFP_{peaks} variance and amplitude were linearly related ($r=0.65$) considering control and unilaterally transected tadpoles together (**Fig. 7C**) and in this context, the reduction between 1/3 and 1/2 of the mirror contralateral glomerular input induced a right-shift compared to control values. Considering that the photoablation procedure left most of OSNs present in the olfactory epithelium intact, the observed displacement suggests that topographical relationships established bilaterally could be relevant to establish a contralaterally driven potentiation. Moreover, our results support that bilateral modulation of glomerular output occurs within a graded scale that is inversely related to the contralateral innervation by OSNs.

Discussion

Odor evoked responses were recorded as changes of the LFP in an olfactory glomerulus labelled in the *Dre.mxn1:GFP X. tropicalis* line (**Fig. 1**) and showed the following characteristics: *i*) were mediated by glutamatergic synapses (**Figs. 2A-C**), *ii*) were triggered by ipsilateral stimulation of OSNs (**Fig. 2F**), and, *iii*) showed an amplitude related to the number of input OSNs (**Fig. 3A**). These characteristics are comparable to responses recorded in the glomerular layer of living rats where negativities are supported by the activation of glutamate receptors and locked to the respiration frequency (Lecoq, Tiret, and Chrapak 2009). Glomerular responses were potentiated by the complete silencing of OSNs projecting to the contralateral olfactory bulb, thus showing the processing of information in the olfactory glomeruli of *Xenopus* tadpoles is not exclusively unilateral and is shaped contralaterally. The observed potentiation was not related to inflammatory mediators associated to injury, because it was caused by a release of the inhibition made by D_2 dopamine receptors present in OSN axon terminals.

The similarities between the recordings obtained (**Fig. 1B**) and those described in rats (Chaigneau et al. 2007), indicate the presence of evolutionary conserved morphological and functional features between both species and therefore, the applicability to vertebrates of the

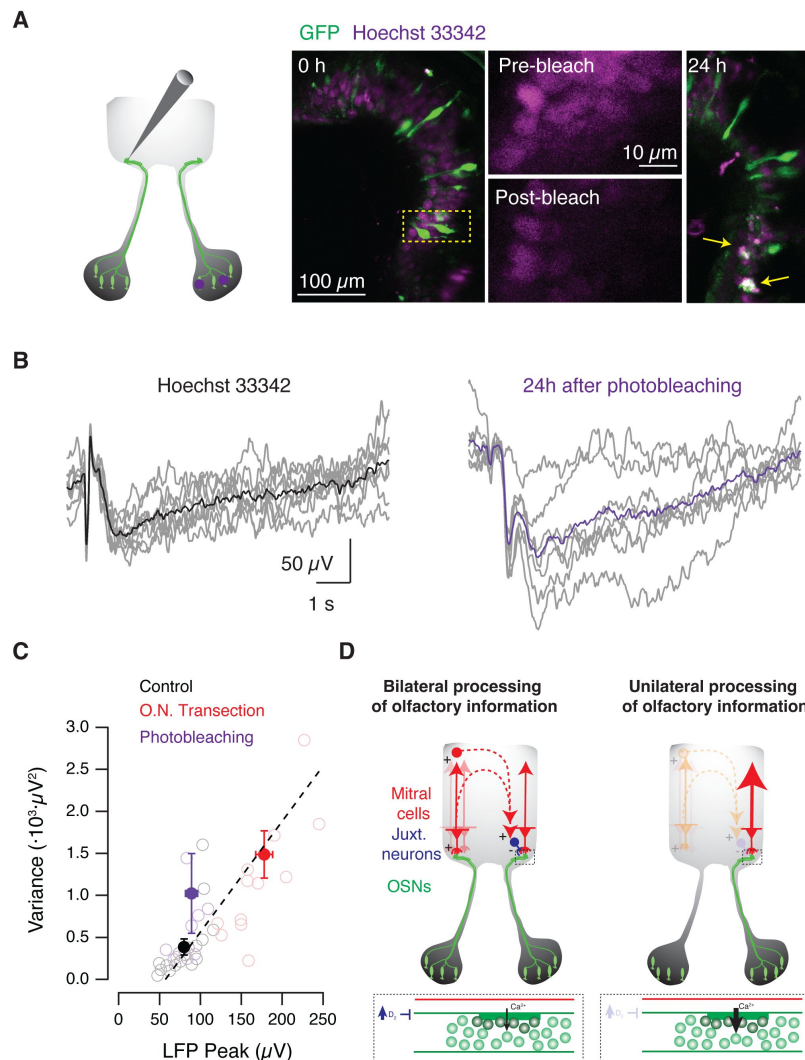


Figure 7.

Changes in odor evoked responses caused by selective photoablation of olfactory sensory neurons innervating the homologous contralateral glomerulus.

A) Odor evoked Local Field Potential (LFP) changes were recorded one day after the selective elimination of olfactory sensory neurons (OSNs) located in the right nasal cavity. After the identification of GFP-positive OSNs in the epithelium labeled with the nuclear marker Hoechst 33342, regions containing ≥ 2 fluorescent neurons were identified and photobleached. Cell targeting was confirmed by the suppression of the nuclear label. Only cells found within the photobleached areas exhibited fragmented nuclei (arrows) 24 hours after photobleaching. **B)** Examples showing odor evoked responses recorded in a tadpole incubated with Hoechst 33342 (black, average), and in a different tadpole 24 hours after the photobleaching of selected regions in the contralateral olfactory epithelium (violet, average). **C)** Plot of variance against the amplitude of LFP_{peaks} in three experimental groups of tadpoles: control ($n=17$), photoablated ($n=10$) and subjected to unilateral olfactory nerve transection ($n=14$). Circles show individual values and dots indicate mean $\pm$ s.e.m. The line shows a fit through values of control and unilaterally transected tadpoles ($r=0.65$). **D)** Model proposed for the bilateral modulation of glomerular output in the olfactory bulb of *Xenopus* tadpoles. Juxtglomerular neurons display tonic dopamine release that inhibits glomerular output by activating presynaptic D2 receptors present in OSNs (dotted box). The constant presence of dopamine within glomeruli is favored by the activity of the contralateral olfactory bulb. When the contribution of the contralateral pathway is suppressed, dopamine release diminishes, and glomerular responses become potentiated. The contralateral modulation of the tonic activity of dopaminergic juxtglomerular neurons corrects for input differences and equalizes the synaptic output of olfactory glomeruli to achieve a bilaterally balanced transfer of information. Axons sent by mitral cells could mediate the pathway described but, the involvement of neurons found in higher brain regions and participating in the processing of olfactory information, such as those of the lateral pallium, could also be considered.

findings here described. Our results achieved using glutamate blockers and uncaging of Rubi-glutamate show that the onset of LFP changes detected in glomeruli is determined by glutamate release from OSNs. The ionic bases supporting the recovery phase of negativities are uncertain but, considering responses obtained by glutamate uncaging were locked to the period defined by the light pulse, the participation of sustained neurotransmitter release is conceivable. This view is also supported by imaging of OMP-synaptotagmin mice, where continuous synaptic vesicle exocytosis occurs over several seconds in OSN axon terminals (Petzold et al. 2008).

The mammalian and amphibian olfactory bulbs display comparable anatomical and cellular organization (Manzini, Schild, and Di Natale 2022) but, it is yet unknown whether the three classes of juxtaglomerular neurons present in rodents: periglomerular, short-axon and external tufted, also exist in *Xenopus*. In mice, short-axon cells are the main source of dopamine to olfactory glomeruli (Kiyokage et al. 2010), however, there is a lack of evidence for such a discrete population in the olfactory bulb of *Xenopus* tadpoles. Dopaminergic neurons are distributed around olfactory glomeruli after NF stage 49 (González et al. 1994) and in adult *Xenopus*, type-I tyrosine hydroxylase positive cells extensively innervate the glomerular layer (Boyd and Delaney 2002). These morphological observations indicate that dopaminergic neurons exist in the glomerular layer of amphibia, but their specific functional properties are unknown. As we did not find evidence for the GABA_B mediated inhibition described in rats (Wachowiak et al. 2005), our results support that dopamine is the main inhibitory neurotransmitter in the glomerular layer of *Xenopus* tadpoles.

To understand how tonic dopamine release in the glomerular layer of *Xenopus* tadpoles is affected by the number of contralateral OSNs, it is necessary to have a comprehensive understanding of olfactory bulb connectivity in *X. tropicalis* tadpoles which is currently lacking. Interneurons of the amphibian olfactory bulb might not display the degree of specialization found in rats or mice, however, there is evidence suggesting that type-1 tyrosine hydroxylase positive interneurons are morphologically similar to external tufted cells present in rodents (Boyd and Delaney 2002). External tufted cells show rhythmic activity (Hayar et al. 2004), are responsible for constructing a map of homologous glomeruli through intrabulbar connections (Lodovichi, Belluscio, and Katz 2003) and act as gatekeepers of glomerular output (Banerjee et al. 2015). Therefore, if the dopaminergic neurons present in the glomerular layer of *Xenopus* tadpoles play analogous roles to external tufted cells, our results call for considering this group of interneurons as candidates to exert the contralateral regulation of glomerular output.

Here, we show that the formation of an odor map in *Xenopus* tadpoles relies on a balanced contribution of the glomerular responses occurring in both olfactory bulbs. Tonic activation of presynaptic D₂ receptors, likely mediated by dopaminergic juxtaglomerular neurons, drives to a constant inhibition of glomerular output. Dopamine release is affected by the number of operative contralateral OSNs so that, in the extreme situation where all mirror olfactory pathways are silenced, dopamine secretion decreases and most of the inhibition is released (Fig. 7D). The consequence is an immediate increase of glomerular responses in the non-damaged pathway, which results in a compensated output signal. A possible direct contralateral targeting by OSNs, as described for the γ -glomerulus present in *X. laevis* tadpoles (Kludt et al. 2015), was ruled because the population labelled in the *Dre.mxn1:GFP* showed exclusively a unilateral distribution. Interhemispheric connections targeting dopaminergic juxtaglomerular neurons could be established by axons of mitral cells as reported in zebrafish (Kermen et al. 2020). However, the participation of neurons of the lateral pallium cannot be disregarded as it is a main center for olfactory processing (Moreno et al. 2008). Either one of these projecting neurons could be targeting contralateral dopaminergic neurons found in the glomerular layer of *X. tropicalis* tadpoles (Fig. 7D). Although the existence of an intraglomerular circuit controlling dopamine release cannot be disregarded (Banerjee et al. 2015), and, considering the differences existing

between olfactory bulb interneurons present in amphibia and mammals (Imamura, Ito, and LaFever 2020 [↗](#)), we modeled the simplest scenario based in only one cell type of juxtaglomerular neuron.

The in vivo evidence presented shows that presynaptic D₂ receptors tonically exert an inhibitory action on calcium buildup in OSN axon terminals. This finding is well aligned with previous observations obtained in mouse brain slices (Ennis et al. 2001 [↗](#)) and describes a relevant biological scenario for the role of dopamine inhibition in the glomerular layer of the olfactory bulb. Both the suppression of contralateral OSN inputs and the local injection of the D₂ antagonist raclopride enhanced glomerular output, a finding that can be explained by the tight coupling existing between calcium influx and neurotransmitter release in OSN axon terminals (Wachowiak et al. 2005 [↗](#)). The likely target of dopamine are N-type voltage gated calcium channels because they are present in presynaptic terminals in olfactory glomeruli (Weiss et al. 2014 [↗](#)) and they can be inhibited through multiple mechanisms activated by D₂ receptors (Kisilevsky and Zamponi 2008 [↗](#)).

Taken altogether our results illustrate a homeostatic mechanism used to compensate a reduced contribution of contralateral OSNs characterized by an enhancement of glomerular output. It is possible that the observed compensation maximizes the role of a pathway relevant to the normal encoding of olfactory information upon transection of an olfactory nerve. Under physiological conditions the contralateral modulation of presynaptic D₂ receptors could be used to correct imbalances in the firing of mitral cells collecting information from bilaterally related glomeruli. Although we can only speculate about such functional role, our lines of evidence show that the mechanism here described is efficiently used for achieving a bilateral compensation of glomerular output in a vertebrate species.

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Reviewer #1 (Public review):

In this study, the authors investigate LFP responses to methionine in the olfactory system of the *Xenopus* tadpole. They show that this response is local to the glomerular layer, arises ipsilaterally, and is blocked by pharmacological blockade of AMPA and NMDA receptors, with little modulation during blockade of GABA-A receptors. They then show that this response is transiently enlarged following transection of the contralateral olfactory nerve, but not the optic lobe nerve. Measurement of ROS- a marker of inflammation- was not affected by contralateral nerve transection, and LFP expansion was not affected by pharmacological blockade of ROS production. Imaging biased towards presynaptic terminals suggests that the enlargement of the LFP has a presynaptic component. A D2 antagonist increases the LFP size and variability in intact tadpoles, while a GABA-B antagonist does not. On this basis, the authors conclude that the increase driven by contralateral nerve transection is due to DA signaling.

Overall, I found the array of techniques and approaches applied in this study to be creatively and effectively employed. However, several of the conclusions made in the Discussion are too strong, given the evidence presented. For example, the authors state that "The observed potentiation was not related to inflammatory mediators associated to injury, because it was caused by a release of the inhibition made by D2 dopamine receptor present in OSN axon terminals." This statement is too strong - the authors have shown that D2 receptors are sufficient to cause an increase in LFP, but not that they are required for the potentiation evoked by nerve transection. The right experiment here would be to get rid of the D2 receptors prior to transection and show that the potentiation is now abolished. In addition, the authors have not shown any data localizing D2 receptors to OSN axon terminals.

Similarly, the authors state, "the onset of LFP changes detected in glomeruli is determined by glutamate release from OSNs." Again, the authors have shown that blockade of AMPA/NMDA receptors decreases the LFP, and that uncaging of glutamate can evoke small negative deflections, but not that the intact signal arises from glutamate release from OSNs. The conclusions about the in vivo contribution of this contralateral pathway are also rather speculative. Acute silencing of one hemisphere would likely provide more insight into the moment-to-moment contributions of bilateral signals to those recorded in one hemisphere.

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Author response:

Thank you for your time and for considering our manuscript as a Reviewed Preprint. We also would like to thank Reviewer 1 for their evaluation of our manuscript.

Here, we present a provisional response to reviewer comments and following their suggestions we will make an effort to: i) increase evidence for the role of dopamine in olfactory glomeruli and ii) delineate the circuit involved mediating the observed potentiation. Next, we briefly describe the set of experiments that are in progress or will be performed to improve our paper.

We will carry out immunostainings for tyrosine hydroxylase to certify that dopamine can be released on the genetically labelled glomerulus. There is a lack of good commercial antibodies for *Xenopus* (we already tried one and did not work, PA1-4679, Thermofisher scientific), but we will look for alternatives. In a previous set of experiments, we attempted to measure dopamine release in the glomerular layer by electroporating olfactory sensory neurons or olfactory bulb neurons with the dopamine sensors dLight1.1 (Addgene #111053) or dLight1.3 (Addgene # 111056). In our hands, fluorescence signals were extremely weak, barely undetectable. Similar results were obtained after electroporating the tectum or the rhombencephalon. We propose to repeat experiments using a more sensitive sensor such as GRAB_DA2m. Other approaches, such as performing single cell transcriptomics of olfactory sensory neurons might be considered to confirm the expression of D2 receptors.

We agree with the reviewer that we should obtain more lines of evidence in support for a presynaptic inhibition mediated by D2 receptors. To gain insight on the bilateral circuit mediating the observed potentiation of glomerular responses we are currently investigating the role of dorsolateral pallium neurons. In *Xenopus* tadpoles the lateral pallium plays an analogous role to the olfactory cortex in amniotes. Preliminary observations show that neurons located in this pallial region respond to ipsilateral stimulation of the olfactory epithelium and if damaged, a contralateral potentiation of glomerular output occurs. We aim to conclude this set of experiments and include it in the paper as we believe it clarifies the circuitry involved.

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